Beacon

Find what you need, right where you are. A better way to find nearby essentials on OpenStreetMap.

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Course	2 MSc AI ML
Subject	FSD CIA 2

Live app: chiragdascodes.github.io/FSDCIA1
GitHub: github.com/Chiragdascodes/FSDCIA1

Part I · Problem Definition and Innovation

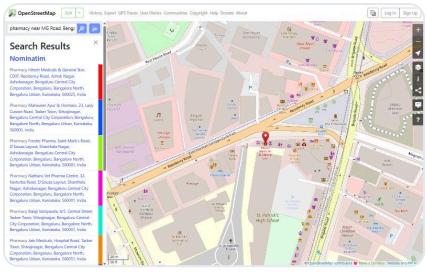
1 (i) Benchmark analysis

I used openstreetmap.org and focused on one workflow: searching for a place, opening it, and getting directions to it. The people I had in mind are newcomers to a city, like students who just moved to Bengaluru and don't know the area yet. They usually need simple things fast: a pharmacy, an ATM, a hospital, a washroom.

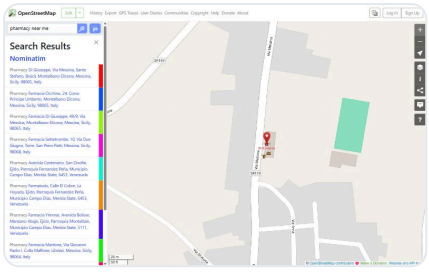
What OSM already has: a search box (Nominatim), a page for every place, "Query features", directions with three routing engines, a share option and map layers. I tested all of these myself. These are the problems I found:

#	What I saw on openstreetmap.org
E1	"pharmacy near MG Road, Bengaluru" works, but every result is a long address line. No distance, no open or closed status, no sorting by nearest.
E2	"pharmacy near me", with the map sitting on Bengaluru, returned pharmacies in Sicily (Italy) and Venezuela. "near me" is just treated as text.
E3	Opening hours are shown as raw tags like `PH,Mo-Su 08:00-23:30`. Nothing tells you if it is open right now.
E4	A place page only offers "Download XML" and "History". No directions, call or share button.
E5	"Query features" mixes useful places with things like "Waste Basket #11117253952" and footpaths. No filters.
E6	From an India-wide view, "pharmacy near MG Road" picked MG Roads in West Bengal and Mumbai, not Bengaluru. It never asks which one you meant.

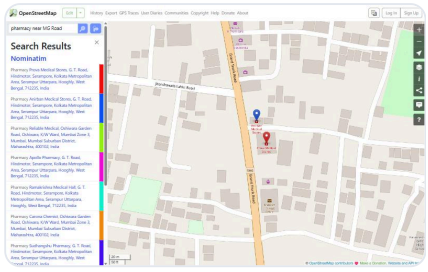
I also checked the data around MG Road. Out of 233 shops and services, only 17 had opening hours. So the app has to be honest when hours are missing.



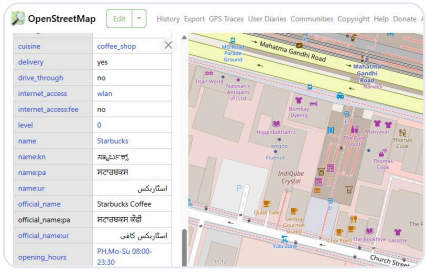
E1 · long addresses, no distance



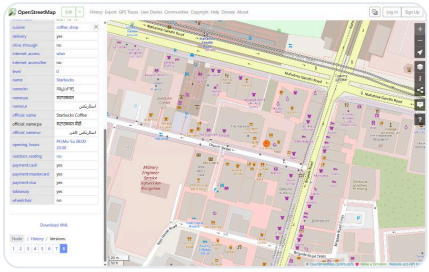
E2 · "near me" goes to Italy



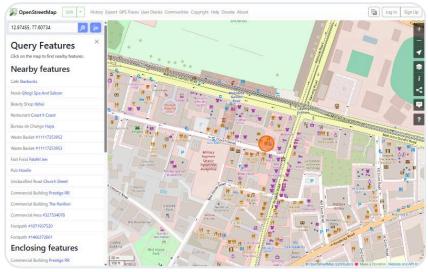
E6 · wrong MG Road



E3 · raw opening hours



E4 · no directions or call



E5 · waste baskets in results

1 (ii) Problem definition

Problem statement: someone new to a city cannot quickly find the nearest essential place that is open right now using OpenStreetMap. "near me" doesn't use their location, street names land in the wrong city, results have no distance or open status, and there is no easy way to get directions from a place.

The impact is real. A student with a fever at night opens several results, reads raw tags, types the address again for directions, and may still end up at a closed shop or in the wrong part of the country. Most people just give up and use another app, even though OSM has the data.

1 (iii) Innovative solution

I built **Beacon**, a web app on top of OpenStreetMap data that does one job well: get you to the nearest essential place.

- It understands normal language: "pharmacy near me", "atm near MG Road", "open 24/7 hospital nearby", even "medical shop" or "petrol bunk".
- "near me" uses your real location. If location is off, it uses your city (Bengaluru by default) and says so.
- If a name is ambiguous, it shows cards like Google Maps with the area, city and distance, nearest first. If one is clearly nearest, it picks it and lets you change.
- Results are sorted by distance with walking time and a plain "Open · closes 23:30" or "Hours unknown".
- Filters for open now, 24/7 and wheelchair, and a radius picker.
- Place card with photo or logo, Call, Website, Save, Share and a big Go Now button.

- Directions for walk, cycle and car, and a Start mode that works like Google Maps navigation.
- Voice search, saved places, a report form for wrong info, a 3D globe, and Liquid Glass styling based on Apple's design guidelines.

Feature	HTML5	Tailwind CSS	JavaScript
Search bar	form role="search", input type="search", combobox list	glass pill, focus ring, responsive width	query parser, debounced suggestions, keyboard nav
Voice search	button with aria-pressed	red mic button, pulse while listening	Web Speech API events
Category chips	buttons in a group, "More" dialog	rounded pills, grid in popover	ResizeObserver fits chips to width
"Which one?" cards	list of buttons	cards with hover state	Photon fetch, distance ranking
Results	ol / li, time	skeleton loaders, badges	Nominatim + Overpass fetch, sorting, filters
Place card	section, table for hours, links	photo header, blue Go Now	opening_hours parsing, Wikidata photo fetch
Directions + Start	ol of steps, dl summary	route card, nav banner	OSRM fetch, GPS watch, route snapping
Report form	form, select, textarea, checkbox, required	error and success styles	validation on blur and submit, localStorage
Layout	header, nav, aside, main sections	sidebar on desktop, bottom sheet on phone	pointer events for the sheet

Part II · JavaScript Implementation

The app uses React with TypeScript for the screens. The core JavaScript the paper asks for is kept in plain `.js` files inside `src/js` so it is easy to read and check.

Meaningful JavaScript events

- **input** on the search box, debounced, fetches live suggestions.
- **submit** on the search form runs the search.
- **click** on map pins opens the place card. Right click on the map shows the address of that spot.
- **keydown** for shortcuts: / to search, Esc to close, L to locate, 1–9 for categories.
- **pointerdown / pointermove / pointerup** to drag the bottom sheet on phones.
- **online / offline** shows a banner when the connection drops.
- Geolocation callbacks and speech recognition **result** and **end** events.

Form with validation and feedback

The "Report a problem" form checks the type, place name, details (at least 15 characters), an optional email or phone, the map location and a consent tick. Each field is checked when you leave it, and everything is checked on submit. Errors show under the field and in a summary at the top. When it is valid you get a success screen and a toast, and the report is saved on the device with an option to post it as an OSM Note. The search box and settings form are validated too.

Fetch API and JSON

- **Photon** for suggestions while typing.
- **Nominatim** for places, nearby category search and reverse geocoding.
- **Overpass** for extra place details.
- **OSRM** for walking, cycling and driving routes.
- **Wikidata** for place photos and brand logos.

Every response is parsed from JSON and turned into one simple place format, so the screens don't care where the data came from.

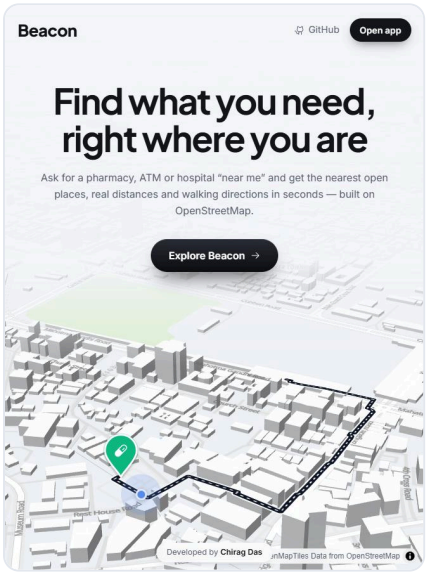
Errors, loading and empty states

- Skeleton loaders and "Finding your location…" while waiting.
- Fast results first from Nominatim, then more details in the background. Overpass was very slow in my tests (29 to 43 seconds), so this matters.

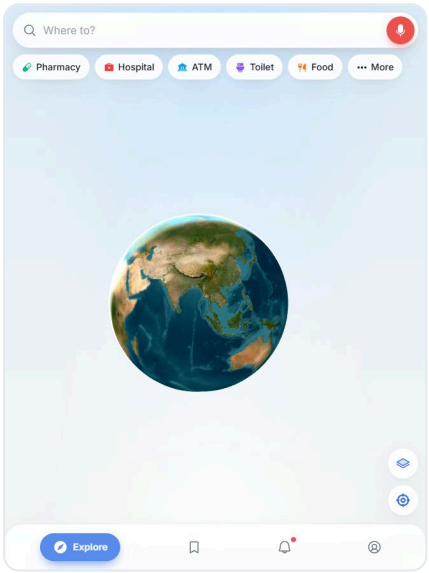
- Every request has a timeout. Errors show a clear message and a Try again button.
- Empty results say so and offer to search a wider area or report a missing place.
- If location is denied, it falls back to the chosen city and tells you.

How it solves the problem

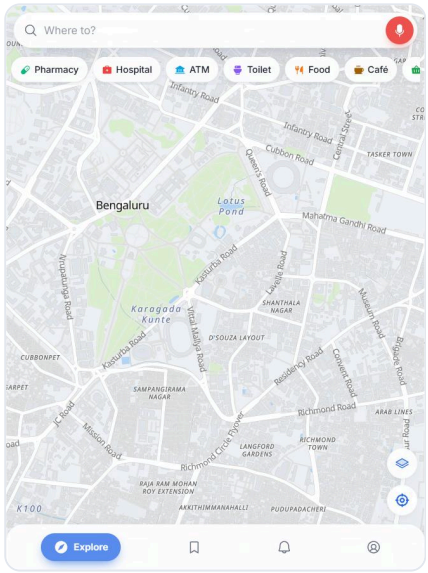
On OpenStreetMap	In Beacon
"near me" goes to Italy (E2)	Uses your location
Wrong MG Road (E6)	Nearest one, or "Which one?" cards
No distance or order (E1)	Nearest first with walking time
Raw hours (E3)	"Open · closes 23:30" and a weekly table
No actions on a place (E4)	Go Now, Call, Website, Save, Share
Noisy results (E5)	Only real places, with filters



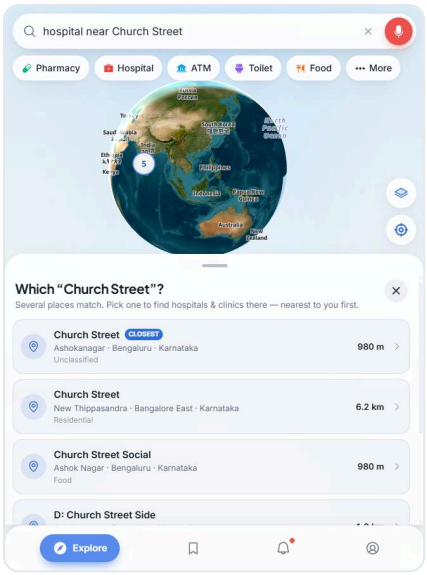
Landing page



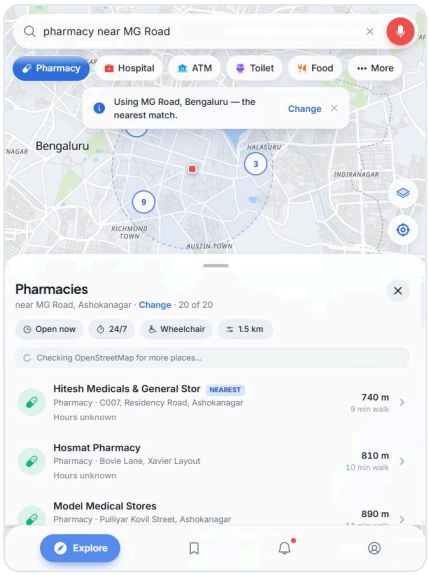
3D globe on start



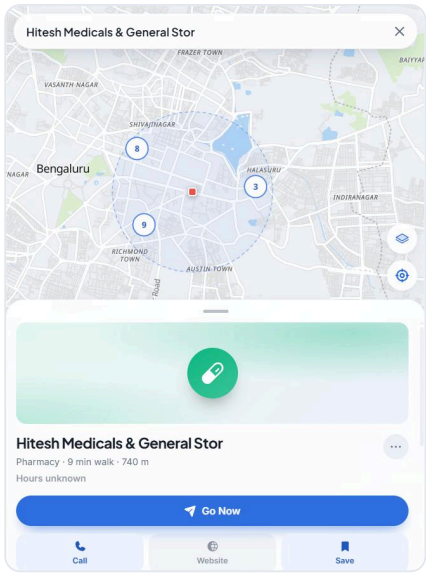
Explore



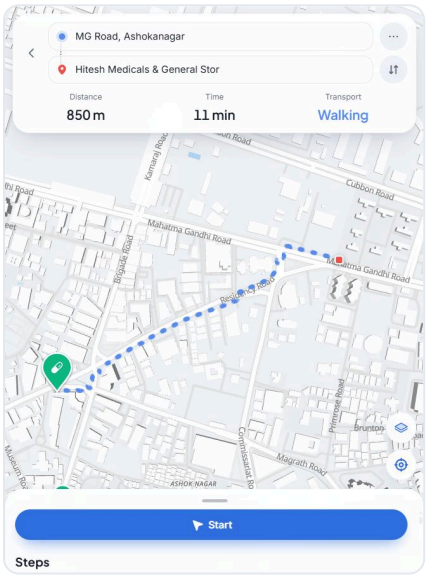
"Which Church Street?"



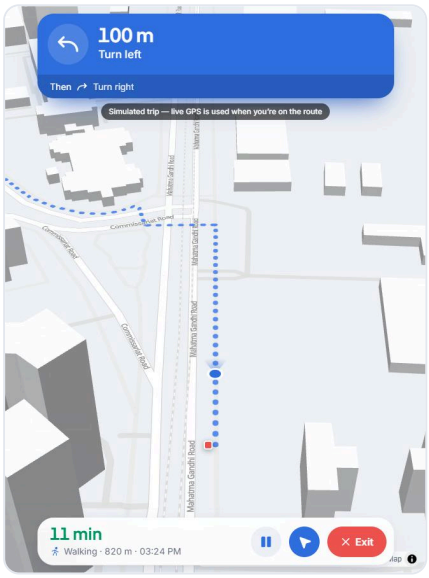
Nearest pharmacies



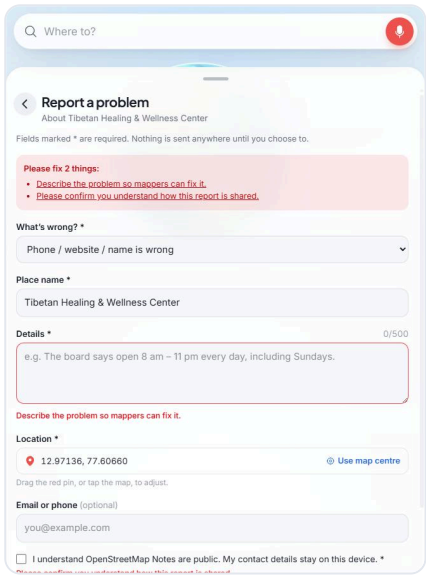
Place card



Route card



Start navigation



Form validation

Part III · Documentation, GitHub and Deployment

Links

Live app: https://chiragdascodes.github.io/FSDCIA1/
GitHub repo: https://github.com/Chiragdascodes/FSDCIA1

The repo has a README and a commit for every step (analysis, setup, core logic, map, search, place card, PWA, redesign, navigation). Every push runs the tests, builds and deploys to GitHub Pages on its own.

Run it locally

`npm install` then `npm run dev`. Tests: `npm test`. Build: `npm run build`.

Tech used

React, TypeScript, Vite, Tailwind CSS, MapLibre GL for the map and globe, OpenFreeMap map styles, Phosphor icons, and free OpenStreetMap services. No login or backend, everything runs in the browser.

Testing

64 automated tests, all passing. They cover the search parser (like "medical shop near me" and "atm near MG Road"), picking the right MG Road, opening hours, distance, form validation, the fetch error handling and the navigation maths. The full log is in `docs/test-results.txt`.

What I tested by hand	Result
"pharmacy near MG Road" from Bengaluru	Picks Bengaluru's MG Road, 20 pharmacies, nearest first
"hospital near Church Street"	Asks which one, shows two Church Streets with area and distance
Location denied	Falls back to Bengaluru and says so
Report form with bad input	Errors shown per field, saved only when valid
Go Now and Start	Route drawn, navigation follows the route
Overpass server down	Results still shown with a small note
Phone and desktop sizes	Layout adapts, no sideways scrolling

Limitations

- Most places in OSM have no opening hours, so many show "Hours unknown".
- The free Overpass and routing servers can be slow or busy.
- No live traffic or public transport routes.
- Navigation needs real GPS to follow you. On a laptop it plays a simulated trip.
- Voice search works in Chrome, Edge and Safari only.
- Reports stay on the device unless you post them as an OSM Note.

AI usage declaration

I used Claude (an AI assistant by Anthropic) while making this project. It helped me analyse openstreetmap.org, write and debug code, and draft this document. I decided the problem, the user group and the design direction, gave the UI references, tested the app myself and asked for changes until it worked the way I wanted. I have gone through the code and can explain how it works.